

IMO(40th) – 1999

First Day

1. Determine all finite sets S at least three points in the plane which satisfy the following condition:
For any two distinct points A and B in S , the perpendicular bisector of the line segment AB is an axis of symmetry for S .
2. Let n be a fixed integer, with $n \geq 2$.

(a) Determine the least constant C such that the inequality $\sum_{1 \leq i < j \leq n} x_i x_j (x_i^2 + x_j^2) \leq C \left(\sum_{1 \leq i \leq n} x_i \right)^4$

holds for all real numbers $x_1, x_2, \dots, x_n \geq 0$.

(b) For this constant C , determine when equality holds.

3. Consider a $n \times n$ square board, where n is fixed even positive integer. The board is divide into n^2 unit squares. We say that two different squares on the board are adjacent if they have a common side.

N unit squares on the board are marked in such a way that every square (marked or unmarked) on the board is adjacent to at least one marked square. Determine the smallest possible value of N .

Second Day

4. Determine all pairs (n, p) of positive integers such that p is prime, n not exceeds $2p$ and $(p-1)^n + 1$ is divisible by n^{p-1} .
5. Two circles G_1 and G_2 are contained inside the circle G , and are tangent to G at the distinct points M and N respectively. G_1 passes through the centre of G_2 . The line passing through the two points of intersection of G_1 and G_2 meets G at A and B . The lines MA and MB meet G_1 at C and D respectively. Prove that CD is tangent to G_2 .
6. Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x - f(y)) = f(f(y)) + xf(y) + f(x) - 1$ for all real numbers x, y .

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